Quiz02/CH1101: Quantum Chemistry & Spectroscopy (Fall Sem 2026/AI, CSE, ICS)

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4. A particle travelling freely in a ring (of radius **r** and with a rotational angle of **ϕ**). The particle has a zero-point energy of (1 Point)

   - ○ $\frac{h^2}{8mr^2\pi^2}$
   - ◉ 0
   - ○ $\frac{4h^2}{8mr^2\pi^2}$

5. The hydrogen-like atomic orbitals, like, 1s, 2s, 2p, etc. are the solutions of the Schrödinger equation. Find the correct orbital for the given wavefunction: (1 Point)

$$\Psi_{2,1,0} = \frac{1}{\sqrt{32\pi}} \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{\frac{Zr}{2a_0}} \left(\frac{Zr}{a_0}\right) \cos\theta$$

- ○ $2p_x$
- ◉ $2p_z$
- ○ $2p_y$

6. Stability of a chemical bond primarily depends on (1 Point)

- ○ anti-bonding molecular orbital
- ◉ bonding molecular orbital

7. From the solutions of the Schrödinger equation for a H-atom, what quantum numbers are obtained? (1 Point)

- ○ n
- ○ l
- ○ m_l
- ◉ All of above

8. Find the bond order for the following homonuclear diatomic molecule: (1 Point)

O_2

- ○ 0.5
- ◉ 2
- ○ 1

9. Assume 8π-electrons of a linear hydrocarbon molecule are modelled by a particle in one-dimensional box model. Choose the correct wavefunction for the ground-state (highest occupied quantum state): (1 Point)

- ○ $\psi_3(x) = \sqrt{\frac{2}{a}} \sin \frac{3\pi x}{a}$
- ◉ $\psi_4(x) = \sqrt{\frac{2}{a}} \sin \frac{4\pi x}{a}$
- ○ $\psi_1(x) = \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a}$

10. A free particle of mass **m** is travelling in one-dimensional box (of length **a**) with infinite potential energy walls. The particle has a zero-point energy of (1 Point)

- ○ 0
- ◉ $\frac{h^2}{8ma^2}$

11. For a particle travelling freely in a ring (of radius **r** and with a rotational angle of **ϕ**), degenerate quantum states exist for: (1 Point)

- ◉ $m_l > 0$
- ○ $m_l = 0$

12. For a homonuclear diatomic molecule the "gerade" symmetry label of a molecular orbital is determined by the (1 Point)

- ◉ center of inversion symmetry operation
- ○ rotation operation
- ○ translation operation

13. The potential energy operator contributes to the Hamiltonian operator for particle in 1-D box model is (1 Point)

- ○ 1
- ◉ zero
- ○ non-zero

14. For a particle travelling freely in a ring (of radius **r** and with a rotational angle of **ϕ**), the Hamiltonian operator is as follows: (1 Point)

- ○ $-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$
- ◉ $-\frac{\hbar^2}{2I}\frac{d^2}{d\phi^2}$

15. For 2pz atomic orbital the nodal plane is on (1 Point)

- ○ xz-plane
- ○ yz-plane
- ◉ xy-plane

16. A particle in 1-D box has the ground state quantum number **n=1**. Choose the correct statement. (1 Point)

- ○ It is more likely that the particle is found near edges.
- ◉ It is more likely that the particle is found near the centre of the box.

17. The time-independent Schrödinger equation for a H-atom has the potential energy V(r): (1 Point)

- ◯ 0
- ◯ 1
- ◉ $-\frac{Ze^2}{4\pi\epsilon_0 r}$

18. The linear combinations of atomic orbitals gives to (1 Point)

- ◯ ψ_+
- ◯ ψ_-
- ◉ Both of above

19. Find the shape of the following hydrogen-like atomic wavefunction/orbital: (1 Point)

$\Psi_{2,0,0}\ (r,\ \theta,\ \phi)$

- ◯ dumbbell-shaped
- ◉ spherical

20. Identify the antibonding molecular orbital: (1 Point)

- ◯ σ_g
- ◉ π_g
- ◯ Both of above

21. Find the correct quantum number of the particle with the following energy: (1 Point)

$\frac{h^2}{8mr^2\pi^2}$

- (●) 1
- ( ) -1
- ( ) Both

22. A benzene molecule containing 6π electrons can be modelled using a particle in ring model for freely moving electrons. Choose the correct wavefunction for the ground-state (highest occupied quantum state): (1 Point)

- ( ) $\Psi_+ = \sqrt{\frac{1}{2\pi}}\, e^{i2\phi}$ and $\Psi_- = \sqrt{\frac{1}{2\pi}}\, e^{-i2\phi}$
- (●) $\Psi_+ = \sqrt{\frac{1}{2\pi}}\, e^{i\phi}$ and $\Psi_- = \sqrt{\frac{1}{2\pi}}\, e^{-i\phi}$
- ( ) $\Psi_+ = \sqrt{\frac{1}{2\pi}}\, e^{i3\phi}$ and $\Psi_- = \sqrt{\frac{1}{2\pi}}\, e^{-i3\phi}$

23. Choose the correct linear combination for generating 2py orbital: (1 Point)

- (●) $\frac{1}{i\sqrt{2}}(\Psi_{2,1,1} - \Psi_{2,1,-1})$
- ( ) $\frac{1}{\sqrt{2}}(\Psi_{2,1,1} + \Psi_{2,1-1})$
- ( ) $\frac{1}{i\sqrt{2}}(\Psi_{2,1,1} + \Psi_{2,1,-1})$

24. State boundary conditions for a particle travelling freely in a ring (of radius **r** and with a rotational angle of **ϕ**) model: (1 Point)

- (●) $\Psi(\phi) = \Psi(\phi + 2\pi)$
- ( ) $\Psi(\phi) \neq \Psi(\phi + 2\pi)$

25. At radial node the radial probability density becomes (1 Point)

- (●) zero

- ◯ non-zero

26. NO+ molecule is isoelectronic to homonuclear diatomic molecule N2 with the following bond order:
(1 Point)

- ◉ 3
- ◯ 1.5
- ◯ 2.5

27. How many radial and angular nodes are present in 3p atomic orbital? (1 Point)

- ◉ 1 and 1
- ◯ 2 and 1
- ◯ 1 and 0
- ◯ 0 and 1

28. For the following angular momentum operator-eigenfunction combination, find the eigenvalue (1 Point)

$-i\hbar\frac{d}{d\phi}$ and $e^{im_l\phi}$

- ◉ $m_l\hbar$
- ◯ $-m_l\hbar$
- ◯ m_l

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